

Team 17 — Pico breadboard wiring reference

MonkMakes Pico BB - Pico with USB at the top, pins in columns c and h - labels at the row ends are exactly what the board prints
Beside the Pico, a hole is named by column + printed label in brackets: a[2] = column a, row printed 2 (GP2). Below the Pico, plain numbers: a26 = column a, row 26.
Inside the Pico: pin name - Pico pin number. The + / - rails on both sides are not used in this layout.

Raspberry Pi 4 header
pin 10 RXD (GPIO15)
pin 8 TXD (GPIO14)
pin 6 GND
male-female wires

GP0 TX → PIR RX
GP1 RX ← Pi TX
G
only for LINK = "UART"

Door contact
MC-38, magnet side
on the lid. Closed
when the lid is shut

wire 1
wire 2

Relay module (5 V)
IN ← GP14, a[14]
GND (DC-) ← b30
VCC (DC+) ← supply +
jumper H or L, see 06

5 V USB supply (relay only)
cut USB cable: red = +, black = -
+ never goes on the breadboard

Wire colours

- Ground (GND)
- Analogue ground (AGND)
- 3.3 V from the Pico
- 5 V relay supply (off board)
- GP16 → green LED
- GP17 → amber LED
- GP18 → red LED
- GP2 → door contact
- Door contact's own 2 wires
- GP26 ← light sensor
- GP15 → buzzer
- GP14 → relay IN
- GP0 TX → Pi pin 10
- GP1 RX ← Pi pin 8

Parts

3 × LED + 330 Ω (+ = long leg)
10 kΩ door pull-up, d26-f26
LDR g26-g27 + 10 kΩ h27-h30
Passive buzzer e29(+)/f29(-)

Before you start

Unplug the Pico USB while
you move wires.

Never connect

5 V to any Pico row.
Rows printed 5V, Vs, 3e, Vr
and Rn: leave empty.

Grounds

Ground rows: 21, 24 and 30
on the left, 29 on the right.
Row 30 right is AGND, only for
the light sensor. It is fed from
[JG] between 28 and 27 = AGND.

Buzzer

Its round body covers the
middle holes of rows 27-30.
Keep them free.

ESP8266 HUZZAH

Own small breadboard, USB
power only. No wire to the
Pico or the Pi — on purpose.

